

[TITLE: concise, keyword-rich, ≤12 words]

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[ABSTRACT PLACEHOLDER]

I. INTRODUCTION

The Krasnoselskii–Mann (KM) iteration [1, 2] is a fundamental relaxation scheme for approximating fixed points of nonexpansive operators. Given an operator T and an initial guess x_0 , the sequence

$$x_{n+1} = (1 - \theta_n) x_n + \theta_n T(x_n), \quad (1)$$

with $\theta_n \in (0, 1)$, converges to a fixed point of T under mild conditions [3]. For a smooth self-map $f : I \rightarrow I$ with constant parameter θ , Eq. (1) defines the θ -transform

$$g_\theta(x) = \theta f(x) + (1 - \theta)x, \quad (2)$$

which preserves every fixed point of f while rescaling their local stability (Sec. II). This paper investigates [STATE FOCUS].

The paper is organised as follows. Section II states the core fixed-point preservation and stability propositions. Section III presents [RESULTS]. Section IV discusses [DISCUSSION]. Section V concludes.

II. THE θ -TRANSFORM: DEFINITIONS AND PROPERTIES

A. Fixed-point preservation and stability scaling

Definition II.1 (θ -transform). Let $f : I \rightarrow I$ be a smooth self-map of an interval $I \subset \mathbb{R}$. For $\theta \in \mathbb{R}$, the θ -transform of f with parameter θ is $g_\theta(x) = \theta f(x) + (1 - \theta)x$.

Proposition II.2 (Fixed-point preservation). *For every $\theta \in \mathbb{R}$, g_θ and f share the same fixed-point set: $g_\theta(x^*) = x^*$ if and only if $f(x^*) = x^*$.*

Proof. $g_\theta(x^*) - x^* = \theta(f(x^*) - x^*)$. For $\theta \neq 0$ this vanishes iff $f(x^*) = x^*$; for $\theta = 0$ the transform collapses to the identity and every point is trivially fixed. $\square$

Proposition II.3 (Stability scaling). *At a fixed point x^* of f ,*

$$g'_\theta(x^*) = 1 - \theta(1 - f'(x^*)). \quad (3)$$

In particular, the derivative vanishes — giving maximally attracting convergence — at the optimal parameter

$$\theta^* = \frac{1}{1 - f'(x^*)}. \quad (4)$$

Proof. Differentiate $g_\theta(x) = \theta f(x) + (1 - \theta)x$: $g'_\theta(x) = \theta f'(x) + (1 - \theta)$. Evaluate at $x = x^*$ and simplify. $\square$

Corollary II.4 (Stability inversion). *A fixed point x^* that is unstable under f (i.e., $|f'(x^*)| > 1$) is stabilised under g_θ for $\theta \in (0, \theta^*)$ when $f'(x^*) > 1$, and for $\theta \in (\theta^*, 0)$ when $f'(x^*) < -1$.*

B. Position-dependent $\theta(x)$

Making θ position-dependent ($\theta \rightarrow \theta(x)$) is the analogue of gauging a global symmetry: the composition rule $\Gamma_\theta \Gamma_\beta = \Gamma_{\theta\beta}$ holds pointwise. The derivative now contains an additional *connection term*:

$$g'(x) = \theta'(x)(f(x) - x) + 1 + \theta(x)(f'(x) - 1). \quad (5)$$

Proposition II.5 (Gauge invariance of fixed points). *The connection term $\theta'(x)(f(x) - x)$ in Eq. (5) vanishes at every fixed point, so the stability formula Eq. (3) holds locally regardless of the profile $\theta(x)$.*

III. [RESULTS — PAPER-SPECIFIC]

A. Model and setup

[Describe the map(s) / system studied. State parameters used. For the logistic map: $f(x) = ax(1 - x)$, $a \in [1, 4]$, $x \in [0, 1]$.]

B. Main result

Theorem III.1 ([NAME]). / [State theorem.]

Proof. $\square$

C. Numerical verification

[Describe computation: Python 3.13, NumPy/SciPy, grid sizes, iteration counts, convergence tolerances. Reference figures: Fig. 1.]

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FIG. 1. **[Panel description.]** [Detailed caption. State all parameter values. Note what each panel/colour/linestyle represents.]

IV. DISCUSSION

[Significance. Comparison to prior work. Limitations. Honest statement of what is proved vs. empirically ob-

served.]

V. CONCLUSIONS

[One paragraph per main contribution. Restate the central result, the method, and the open questions it raises.]

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Appendix A: Numerical Methods and Software

All computations use Python 3.13 with NumPy [4], SciPy [5], and Matplotlib [6]. [Describe: grid sizes, iteration counts, convergence criteria, random seeds, Dask parallelisation where applicable.]

Appendix B: [ADDITIONAL PROOFS]

[Move lengthy derivations here to keep the main text readable.]

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- [1] M. A. Krasnoselskiĭ, Two remarks on the method of successive approximations, *Uspekhi Matematicheskikh Nauk* **10**, 123 (1955), in Russian. The paper introduces the relaxed iteration $x_{n+1} = \frac{1}{2}(x_n + Tx_n)$ for nonexpansive T , the progenitor of the K-M family.
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 - [4] C. R. Harris *et al.*, Array programming with NumPy, *Nature* **585**, 357 (2020).
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